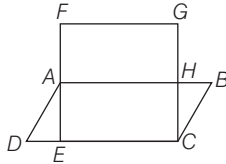
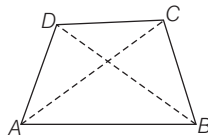


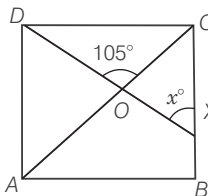
- 6.** If $ABCD$ is a rhombus, then
 (a) $AC^2 + BD^2 = 4AB^2$ (b) $AC^2 + BD^2 = AB^2$
 (c) $AC^2 + BD^2 = 2AB^2$ (d) $2(AC^2 + BD^2) = 3AB^2$
- 7.** A point O in the interior of a rectangle $ABCD$ is joined with each of the vertices A, B, C and D . Then,
 (a) $OB + OD = OC + OA$ (b) $OB^2 + OA^2 = OC^2 + OD^2$
 (c) $OB \cdot OD = OC \cdot OA$ (d) $OB^2 + OD^2 = OC^2 + OA^2$
- 8.** In a trapezium $ABCD$, if $AB \parallel CD$, then $AC^2 + BD^2$ is equal to
 (a) $BC^2 + AD^2 + 2AB \cdot CD$ (b) $AB^2 + CD^2 + 2AD \cdot BC$
 (c) $AB^2 + CD^2 + 2AB \cdot CD$ (d) $BC^2 + AD^2 + 2BC \cdot AD$
- 9.** $ABCD$ is a trapezium in which $AB \parallel CD$ and $AB = 2CD$. Its diagonals intersect each other at O , then the ratio of the areas of the $\triangle AOB$ and $\triangle COD$ is
 (a) 1:2 (b) 2:1 (c) 4:1 (d) 1:4
- 10.** In the figure, $ABCD$ is a parallelogram with $AD = a$ units, $DC = 2a$ units and $DE : EC = 1 : 2$. $CEFG$ is a rectangle with $FE = 3AE$. What is the ratio of the area of a parallelogram and the rectangle?



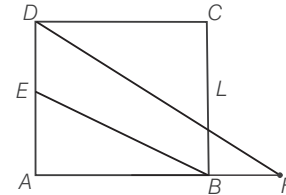
- (a) 1:1 (b) 1:2 (c) $\frac{a^2}{5} : 1$ (d) 2:1
- 11.** In the adjoining figure, $ABCD$ is a quadrilateral in which AB is the longest side and CD is the shortest side, then



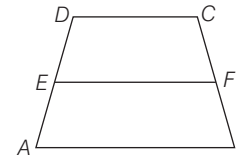
- (a) $\angle C > \angle A$ and $\angle D > \angle B$ (b) $\angle C > \angle A$ and $\angle B > \angle D$
 (c) $\angle C < \angle A$ and $\angle D < \angle B$ (d) $\angle C < \angle A$ and $\angle D = \angle B$
- 12.** In the given figure, $ABCD$ is a square. A line segment DX cuts the side BC at X and the diagonal AC at O such that $\angle COD = 105^\circ$ and $\angle OXC = x$. The value of x is



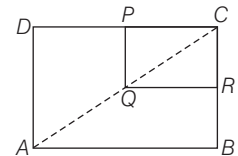
- 13.** In the given figure, $ABCD$ is a parallelogram and E is the mid-point of AD . A line through D , drawn parallel to EB , meets AB produced at F and BC at L . Then,



- (a) $AF = 2DC$ (b) $2AF = 3DC$
 (c) $2DF = 3DL$ (d) $AF + DF = 3DC + DL$
- 14.** Let $ABCD$ be a trapezium in which $AB \parallel DC$ and let E be the mid-point of AD . Let F be a point on BC such that $EF \parallel AB$. Then, consider the following statements.



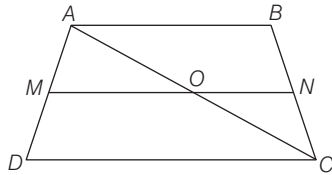
- (a) F is the mid-point of BC (b) $EF = \frac{1}{2}(AB + DC)$
 (c) Both (a) and (b) (d) None of these
- 15.** In the figure given below, $ABCD$ and $PQRC$ are rectangles, where Q is the mid-point of AC , then



- (a) $DP = PC$ (b) $\frac{1}{3}PR = \frac{3}{2}DB$
 (c) $DP = PQ$ (d) $DP = \frac{1}{2}AC$
- 16.** How many diagonals are there in a octagon?
 (a) 8 (b) 16 (c) 18 (d) 20
- 17.** The angles of a pentagon are in the ratio 1:2:3:5:9, the largest angle is
 (a) 81° (b) 135° (c) 243° (d) 249°
- 18.** The ratio of an interior angle to the exterior angle of a regular polygon is 5 : 1. The number of sides of polygon is
 (a) 10 (b) 11 (c) 12 (d) 14
- 19.** The measure of each angle of a regular hexagon is
 (a) 90° (b) 120° (c) 105° (d) 135°
- 20.** Each interior angle of a regular polygon is 150° . The number of sides of the polygon is
 (a) 4 (b) 8 (c) 12 (d) 16

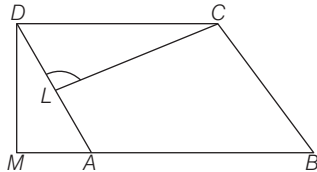
- 21.** Let $LMNP$ be a parallelogram and NR be perpendicular to LP . If the area of the parallelogram is six times the area of $\triangle RNP$ and $RP = 6$ cm what is LR equal to?
 (a) 15 cm (b) 12 cm (c) 9 cm (d) 8 cm

- 22.** $ABCD$ is a trapezium in which $AB \parallel CD$. M and N are the mid-points of AD and BC , respectively. If $AB = 12$ cm and $MN = 14$ cm find CD .



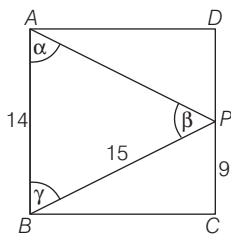
- (a) 2 cm (b) 5 cm (c) 12 cm (d) 16 cm

- 23.** $ABCD$ is a parallelogram. CL is perpendicular to AD and DM is perpendicular to BA produced. If $CD = 16$ units, $DM = 12$ units and $CL = 15$ units then $AD = ?$



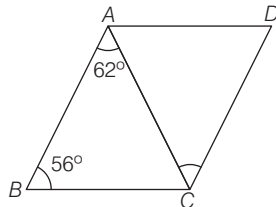
- (a) 12.8 units (b) 13.6 units (c) 11.1 units (d) 12.4 units

- 24.** $ABCD$ is a rectangle. $PC = 9$ cm, $BP = 15$ cm, $AB = 14$ cm. Then, the angles of $\triangle APB$ are such that



- (a) $\alpha > \beta > \gamma$ (b) $\alpha > \gamma > \beta$ (c) $\beta > \gamma > \alpha$ (d) $\alpha < \beta < \gamma$

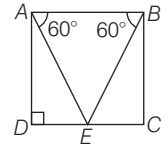
- 25.** $ABCD$ is a rhombus with $\angle ABC = 56^\circ$ and $\angle BAC = 62^\circ$, then $\angle ACD$ is equal to



- (a) 90° (b) 60° (c) 56° (d) 62°

- 26.** One angle of a pentagon is 140° . If the remaining angles are in the ratio $1 : 2 : 3 : 4$, then the size of the greatest angle is
 (a) 150° (b) 180° (c) 160° (d) 170°

- 27.** In the given figure $ABCD$ is a quadrilateral with AB parallel to DC and AD parallel to BC , $\angle ADC$ is a right angle. If the perimeter of the $\triangle ABE$ is 6 units, what is the area of the quadrilateral?

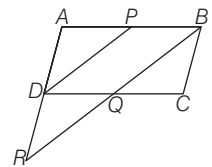


- (a) $2\sqrt{3}$ sq units (b) 4 sq units
 (c) 3 sq units (d) $4\sqrt{3}$ sq units

- 28.** A square and a rhombus have the same base and the rhombus is inclined at 30° . What is the ratio of area of the square to the area of the rhombus?

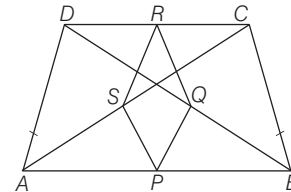
- (a) $\sqrt{2} : 1$ (b) $2 : 1$ (c) $1 : 1$ (d) $2 : \sqrt{3}$

- 29.** P is the mid-point of side AB of parallelogram $ABCD$. A line through B parallel to PD meets DC at Q and AD produced at R . Then BR is equal to



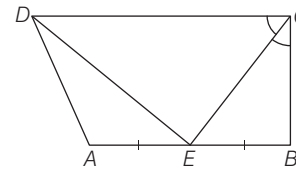
- (a) BQ (b) $\frac{1}{2}$ (c) $2BQ$ (d) None of these

- 30.** $ABCD$ is a trapezium in which $AB \parallel DC$ and $AD = BC$. If P, Q, R, S be respectively the mid-point of BA, BD, CD and CA , then $PQRS$ is a



- (a) rhombus (b) rectangle
 (c) parallelogram (d) square

- 31.** $ABCD$ is a parallelogram, E is the mid-point of AB and CE bisects $\angle BCD$. Then $\angle DEC$ is



- (a) 60° (b) 90° (c) 100° (d) 120°

- 32.** $ABCD$ is a square, P, Q, R and S are points on the sides AB, BC, CD and DA , respectively such that $AP = BQ = CR = DS$. What is the value of $\angle SPQ$?

- (a) 30° (b) 45° (c) 60° (d) 90°

- 33.** The middle points of the parallel sides AB and CD of a parallelogram $ABCD$ are P and Q , respectively. If AQ and CP divide the diagonal BD into three parts BX, XY and YD , then which one of the following is correct?

- (a) $BX \neq XY \neq YD$ (b) $BX = YD \neq XY$
 (c) $BX = XY = YD$ (d) $XY = 2BX$

- 34.** A parallelogram and a rectangle stand on the same base and on the same side of the base with the same height. If l_1, l_2 be the perimeters of the parallelogram and the rectangle respectively, then which one of the following is correct?

(a) $l_1 < l_2$ (b) $l_1 = l_2$
(c) $l_1 > l_2$ but $l_1 \neq 2l_2$ (d) $l_1 = 2l_2$

- 35.** Two similar parallelograms have corresponding sides in the ratio $1:k$. What is the ratio of their areas?

(a) $1:3k^2$ (b) $1:4k^2$ (c) $1:k^2$ (d) $1:2k^2$

- 36.** Let $WXYZ$ be a square. Let P, Q and R be the mid-points of WX, XY and ZW , respectively and K, L be the mid-points of PQ and PR , respectively. What is the value of

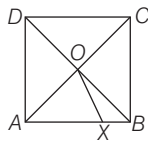
$\frac{\text{area of } \triangle PKL}{\text{area of square } WXYZ}$?

(a) $\frac{1}{32}$ (b) $\frac{1}{16}$
(c) $\frac{1}{8}$ (d) $\frac{1}{64}$

- 37.** ABC is a triangle in which $AB = AC$. Let BC be produced to D . From a point E on the line AC let EF be a straight line such that EF is parallel to AB . Consider the quadrilateral $ECDF$ thus formed. If $\angle ABC = 65^\circ$ and $\angle EFD = 80^\circ$, then what is the value of $\angle FDC$?

(a) 43° (b) 41°
(c) 37° (d) 35°

- 38.** In the given figure, $ABCD$ is a square in which $AO = AX$. What is $\angle XO B$?

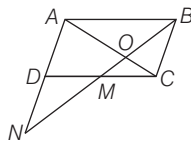


(a) 22.5° (b) 25°
(c) 30° (d) 45°

- 39.** Two sides of a parallelogram are 10 cm and 15 cm. If the altitude corresponding to the side of length 15 cm is 5 cm, then what is the altitude to the side of length 10 cm?

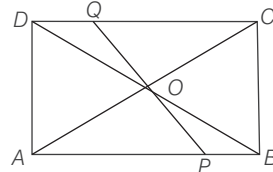
(a) 5 cm (b) 7.5 cm
(c) 10 cm (d) 15 cm

- 40.** In the given figure, M is the mid-point of the side CD of the parallelogram $ABCD$. What is $ON : OB$?



(a) $3:2$ (b) $2:1$ (c) $3:1$ (d) $5:2$

- 41.** $ABCD$ is a parallelogram whose diagonals AC and BD intersect at O . A line segment PQ through O meets AB at P and DC at Q .



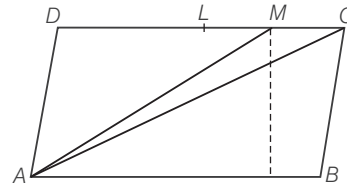
(a) $\text{ar}(\text{quadrilateral } APQD) = \frac{1}{2} \text{ar}(\text{gm } ABCD)$

(b) $\text{ar}(\text{quadrilateral } APQD) = \frac{1}{4} \text{ar}(\text{gm } ABCD)$

(c) $\text{ar}(\text{quadrilateral } APQD) = \frac{1}{3} \text{ar}(\text{gm } ABCD)$

(d) None of the above

- 42.** In the figure, $ABCD$ is a $\parallel\text{gm}$, L is the mid-point of DC and M is the mid-point of LC . If $\text{ar}(\triangle AMC) = 4 \text{ cm}^2$, then $\text{ar}(\parallel\text{gm } ABCD)$ is



(a) 32 cm^2 (b) 16 cm^2
(c) 64 cm^2 (d) Data insufficient

- 43.** Consider the following statements.

I. All squares are not parallelograms.

II. All parallelograms are trapezium.

III. All squares are rhombuses and also rectangles.

IV. All rhombuses are parallelograms.

Which of the above statements are correct?

(a) I and II (b) III and IV (c) I and III (d) II and IV

- 44.** Consider the following statements.

I. If the side of a rhombus is 10 cm. Its diagonals should have values 16 cm and 12 cm.

II. The diagonals of a rhombus cut at right angles.

III. The diagonals do not bisect each other.

Which of the above statements are true?

(a) I and II (b) II and III (c) I and III (d) I, II and III

- 45.** Consider the following statements in respect of a quadrilateral

I. The line segments joining the mid-points of the two pairs of opposite sides bisect each other at the point of intersection.

II. The area of the quadrilateral formed by joining the mid-points of the four adjacent sides is half of the total area of the quadrilateral.

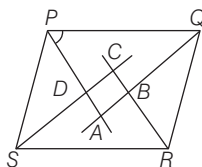
Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

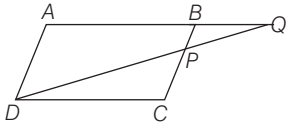
> PREVIOUS YEARS' QUESTIONS

Directions (Q.Nos. 46-47) Let $ABCD$ be a quadrilateral, the diagonals AC and BD meet at O and the perpendicular drawn from A to CD , meet CD at E . Further, $AO : OC = BO : OD$, $AB = 30$ cm, $CD = 40$ cm and the area of the quadrilateral $ABCD$ is 1050 cm^2 .

- 46.** What is the value of BE ? ☑ 2012 I
 (a) 30 cm (b) $30\sqrt{2}$ cm
 (c) $30\sqrt{3}$ cm (d) None of these
- 47.** What is the area of the $\triangle ADC$? ☑ 2012 I
 (a) 300 cm^2 (b) 450 cm^2
 (c) 600 cm^2 (d) None of these
- 48.** $ABCD$ is a rhombus with diagonals AC and BD . Then, which one among the following is correct? ☑ 2012 I
 (a) AC and BD bisect each other but not necessarily perpendicular to each other
 (b) AC and BD are perpendicular to each other but not necessarily bisect each other
 (c) AC and BD bisect each other and perpendicular to each other
 (d) AC and BD neither bisect each other nor perpendicular to each other
- 49.** The sides of a parallelogram are 12 cm and 8 cm long and one of the diagonals is 10 cm long. If d is the length of other diagonal, then which one of the following is correct? ☑ 2012 I
 (a) $d < 8$ cm (b) $8 \text{ cm} < d < 10$ cm
 (c) $10 \text{ cm} < d < 12$ cm (d) $d > 12$ cm
- 50.** Let X be any point within a square $ABCD$. On AX a square $AXYZ$ is described such that D is within it. Which one of the following is correct? ☑ 2012 II
 (a) $AX = DZ$ (b) $\angle ADZ = \angle BAX$
 (c) $AD = DZ$ (d) $BX = DZ$
- 51.** $ABCD$ is a parallelogram. If the bisectors of the $\angle A$ and $\angle C$ meet the diagonal BD at points P and Q respectively, then which one of the following is correct? ☑ 2012 II
 (a) $PCQA$ is a straight line (b) $\triangle APQ$ is similar to $\triangle PCQ$
 (c) $AP = CP$ (d) $AP = AQ$
- 52.** The locus of a point in rhombus $ABCD$ which is equidistant from A and C is ☑ 2012 II
 (a) a fixed point on diagonal BD
 (b) diagonal BD (c) diagonal AC
 (d) None of the above
- 53.** In the figure given below, $PQRS$ is a parallelogram. If AP , AQ , CR and CS are bisectors of $\angle P$, $\angle Q$, $\angle R$ and $\angle S$ respectively, then $ABCD$ is a ☑ 2013 I
 (a) square (b) rhombus
 (c) rectangle (d) None of these



- 54.** $ABCD$ is a trapezium with parallel sides $AB = 2$ cm and $DC = 3$ cm. E and F are the mid-points of the non-parallel sides. The ratio of area of $ABFE$ to area of $EFCD$ is ☑ 2013 I
 (a) $9 : 10$ (b) $8 : 9$ (c) $9 : 11$ (d) $11 : 9$
- 55.** If the diagonals of a rhombus are 4.8 cm and 1.4 cm, then what is the perimeter of the rhombus? ☑ 2013 II
 (a) 5 cm (b) 10 cm (c) 12 cm (d) 20 cm
- 56.** $ABCD$ is a quadrilateral such that $BC = BA$ and $CD > AD$. Which one of the following is correct? ☑ 2013 II
 (a) $\angle BAD = \angle BCD$ (b) $\angle BAD < \angle BCD$
 (c) $\angle BAD > \angle BCD$ (d) $2\angle BAD = \angle BCD$
- 57.** Let $ABCD$ be a parallelogram. Let P , Q , R and S be the mid-points of sides AB , BC , CD and DA , respectively. Consider the following statements.
 I. Area of triangle $APS <$ Area of triangle DSR , if $BD < AC$.
 II. Area of triangle $ABC = 4$ (Area of triangle BPQ).
 Select the correct answer using the codes given below ☑ 2014 I
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 58.** In a trapezium, the two non-parallel sides are equal in length, each being of 5 cm. The parallel sides are at a distance of 3 cm apart. If the smaller side of the parallel sides is of length 2 cm, then the sum of the diagonals of the trapezium is ☑ 2014 I
 (a) $10\sqrt{5}$ cm (b) $6\sqrt{5}$ cm (c) $5\sqrt{5}$ cm (d) $3\sqrt{5}$ cm
- 59.** Two light rods $AB = a + b$, $CD = a - b$ symmetrically lying on a horizontal plane. They are kept intact by two strings AC and BD . The perpendicular distance between rods is a . The length of AC is given by ☑ 2014 I
 (a) a (b) b (c) $\sqrt{a^2 - b^2}$ (d) $\sqrt{a^2 + b^2}$
- 60.** Let $ABCD$ be a parallelogram. Let X and Y be the mid-points of the sides BC and AD , respectively. Let M and N be the mid-points of the sides AB and CD , respectively. Consider the following statements.
 I. The straight line MX cannot be parallel to YN .
 II. The straight lines AC , BD , XY and MN meet at a point.
 Which of the statement(s) given above is/are correct? ☑ 2014 II
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 61.** If each interior angle of a regular polygon is 135° , then the number of diagonals of the polygon is equal to ☑ 2015 I
 (a) 54 (b) 48 (c) 20 (d) 18

- 62.** If a star figure is formed by elongating the sides of a regular pentagon, then the measure of each angle at the angular points of the star figure is $\checkmark$ 2015 I
(a) 36° (b) 35° (c) 32° (d) 30°
- 63.** If X is any point within a square $ABCD$ and on AX a square $AXYZ$ is described, which of the following is/are correct?
I. $BX = DZ$ or $BZ = DX$
II. $\angle ABX = \angle ADZ$ or $\angle ADX = \angle ABZ$
Select the correct answer using the code given below. $\checkmark$ 2015 II
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 64.** In the adjoining figure, $ABCD$ is a parallelogram. P is a point on BC such that $PB:PC = 1:2$. DP and AB when both produced meet at Q . If area of $\triangle BPQ$ is 20 sq units, the area of $\triangle DCP$ is $\checkmark$ 2015 I
(a) 20 sq units (b) 30 sq units
(c) 40 sq units (d) None of these
- 
- 65.** $ABCD$ is a parallelogram with AB and AD as adjacent sides. If $\angle A = 60^\circ$ and $AB = 2AD$, then the diagonal BD will be equal to $\checkmark$ 2015 II
(a) $\sqrt{2}AD$ (b) $\sqrt{3}AD$ (c) $2AD$ (d) $3AD$
- 66.** If each interior angle of a regular polygon is 140° then number of vertices of the polygon is equal to $\checkmark$ 2016 I
(a) 10 (b) 9 (c) 8 (d) 7
- 67.** Consider the following statements.
I. There exists a regular polygon whose exterior angle is 70° .
II. Let $n \geq 5$, then the exterior angle of any regular polygon of n sides is acute.
Which of the above statements is/are correct? $\checkmark$ 2016 I
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 68.** Consider the following statements.
I. If $n \geq 3$ and $m \geq 3$ are distinct positive integers, then the sum of the exterior angles of a regular polygon of m sides is different from the sum of the exterior angles of a regular polygon of n sides.
II. Let m, n be integers such that $m > n \geq 3$. Then, the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides and their sum is $(m+n)\pi/2$.
Which of the above statements is/are correct? $\checkmark$ 2016 I
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

ANSWERS

1	c	2	b	3	c	4	b	5	d	6	a	7	d	8	a	9	c	10	b
11	a	12	b	13	a	14	b	15	a	16	d	17	c	18	c	19	b	20	c
21	b	22	d	23	a	24	a	25	d	26	c	27	a	28	b	29	c	30	a
31	b	32	d	33	c	34	c	35	c	36	b	37	d	38	a	39	b	40	b
41	a	42	a	43	b	44	a	45	c	46	b	47	c	48	c	49	d	50	d
51	b	52	a	53	c	54	c	55	b	56	c	57	b	58	b	59	d	60	b
61	c	62	a	63	c	64	d	65	b	66	b	67	b	68	d				

HINTS AND SOLUTIONS

- 1.** (c) As $\angle A + b^\circ = 180^\circ$ [linear pair]
 $\Rightarrow \angle A = 180^\circ - b$
 Also, $\angle C + a^\circ = 180^\circ$ [linear pair]
 $\Rightarrow \angle C = 180^\circ - a^\circ$
 But $\angle A + \angle B + \angle C + \angle D = 360^\circ$
 [by angle sum property of a quadrilateral]
 $\Rightarrow (180^\circ - b^\circ) + x^\circ + (180^\circ - a^\circ) + y^\circ = 360^\circ$
 $\Rightarrow x^\circ + y^\circ = a^\circ + b^\circ$
- 2.** (b) $\because \angle 1 = \frac{1}{2} \angle C$ and $\angle 2 = \frac{1}{2} \angle D$
 In $\triangle DEC$, $\angle 1 + \angle 2 + \angle CED = 180^\circ$
 [angle sum property of a triangle]
 $\therefore \angle CED = 180^\circ - (\angle 1 + \angle 2)$
 In quadrilateral $ABCD$,
 $\angle A + \angle B + \angle C + \angle D = 360^\circ$
 $\angle A + \angle B + 2(\angle 1 + \angle 2) = 360^\circ$
 $\angle A + \angle B = 360^\circ - 2(\angle 1 + \angle 2)$
 $\angle A + \angle B = 2[180^\circ - (\angle 1 + \angle 2)]$
 $\angle A + \angle B = 2 \angle CED$
- 3.** (c) In $\triangle ABC$, $\triangle ACD$, $\triangle BCD$ and $\triangle ABD$
 $AB + BC > AC$, $CD + DA > AC$
 $BC + CD > BD$, $DA + AB > BD$
 [Sum of any two sides is greater than the third side.]
 adding above inequalities,
 $2(AB + BC + CD + DA) > 2(AC + BD)$
 $(AB + BC + CD + DA) > (AC + BD)$
- 4.** (b) Let h be the height of the parallelogram. Then, clearly $h < q$
 $\therefore R = p \times h < p \times q = S$
 $R < S$